

 AGENTIC-RAG-Ω — Marketing Agent (Design + Publish)

n8n 6 :

PLAN → RETRIEVE → SYNTHESIZE → GENERATE IMAGE → GENERATE VIDEO PROMPT → PUBLISH → MEMORY

1. Retrieval Planning

2. Retrieve

3. Filter/Rerank/Grade

4. AI Synthesize

5. Generate Image

6. Prepare Video

7. Publish

8. Save Memory
- + +

5

+ caption + hashtags + prompts

DALL-E 3

prompt (Runway/Pika)

Instagram / Facebook / LinkedIn / X / TikTok



AGENTIC_RAG_OMEGA_marketing_agent_design_publish.json —23 node

ag_rag_omega_schema.sql



1.

psql -U postgres -c "CREATE DATABASE marketing_engine;"

psql -U postgres -d marketing_engine -f ag_rag_omega_schema.sql

2. Workflow

- n8n → Workflows → Import from File
- AGENTIC_RAG_OMEGA_marketing_agent_design_publish.json

3. Credentials

Credential		
OpenAI API	openAiApi	+ DALL-E
PostgreSQL	postgres	+
Meta Access Token	httpHeaderAuth	Instagram + Facebook
LinkedIn Access Token	httpHeaderAuth	LinkedIn

Credential		
X/Twitter OAuth2	OAuth2	X/Twitter

n8n → **Settings** → **Variables**:

```
INSTAGRAM_BUSINESS_ACCOUNT_ID = your_ig_business_account_id
FACEBOOK_PAGE_ID = your_fb_page_id
LINKEDIN_PERSON_URN = your_linkedin_person_urn
```



```
curl -X POST https://YOUR_N8N_URL/webhook/marketing-agent -H "Content-Type: application/json" -d '{
  "task": "📢📢📢📢 📢📢📢📢 📢📢📢📢 📢📢📢📢",
  "brand": "JuicyCo",
  "product": "📢📢📢📢 📢📢📢📢📢 📢📢📢📢",
  "audience": "📢📢📢📢 18-30 📢📢📢📢📢 📢📢📢📢📢",
  "platform": "instagram",
  "contentType": "image",
  "tone": "fun",
  "publish": true,
  "sessionId": "sess-001"
}'
```

```
{
  "success": true,
  "session_id": "sess-001",
  "content_brief": {
    "strategy": "0000 0000 0000 00 000000 "...000000000",
    "visual_concept": "0000 000000 0000 00 00000 000 000000 0000 "...0000",
    "caption": "$$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]{/usr/share/docminer/emoji/
      ↳ 2600-fe0f.png}}}}$$ 0000 0000 !000000000
      ↳ $$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]{/usr/share/docminer/emoji/
      ↳ 1f34a.png}}}}$$ 0000 100% 00000 0000 000 "...0000",
    "hashtags": ["#_0000"00000, "#_0002026", "#"0000, "#"0000, "#"000000],
    "cta": "0000 0000 0000000 !0000000000
  },
  "generated_assets": {
    "image_url": "https://oaidalleapiprodscus.blob.core.windows.net/...",
    "video_prompt": "A cinematic shot of fresh oranges being squeezed...",
    "video_status": "pending_external_api"
  },
  "publish_status": {
```

```
"platform": "instagram",
"published": true,
"post_id": "179xxxxx"
},
"confidence": 0.88
}
```



(DALL-E 3)

prompt DALL-E 3.

API . prompt :

- 1. Runway ML (<https://api.runwayml.com>)
- 2. Pika Labs
- 3. Kling AI
- 4. prompt



Instagram	✓	Meta Business Account + Access Token
Facebook	✓	Facebook Page + Access Token
LinkedIn	✓	LinkedIn Developer App + Access Token
X/Twitter	✓	OAuth2 App
TikTok	⚠ Draft	Creator Portal API)



API Keys

Meta (Instagram + Facebook)

- 1. [Meta for Developers](#)
- 2. App → “Business”
- 3. Access Token) instagram_basic, instagram_content_publish, pages_manage_posts)
- 4. Instagram Business Account ID Facebook Business Manager

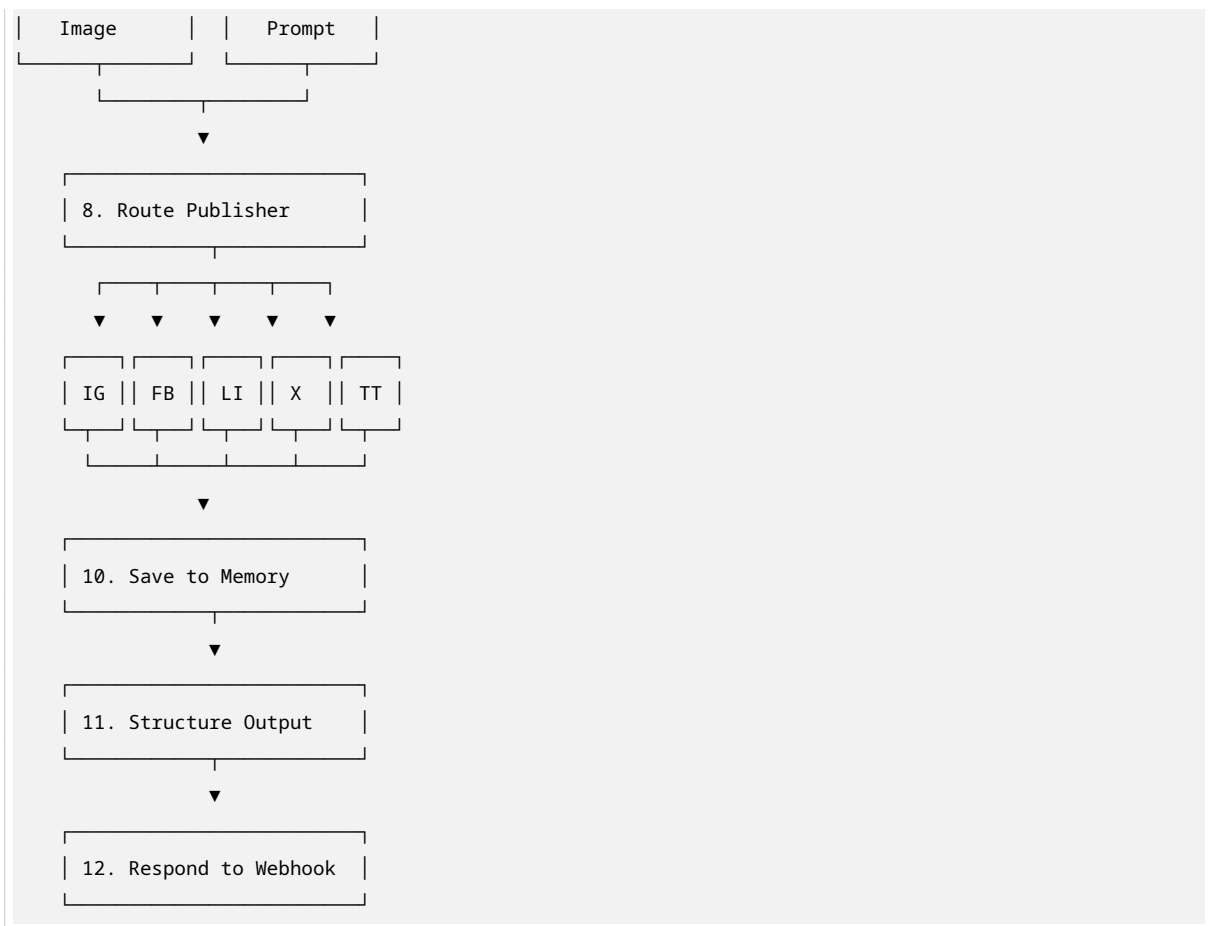
LinkedIn

1. [LinkedIn Developers](#)
2. App → “Share on LinkedIn”
3. **Access Token**) w_member_social)
4. **Person URN** https://api.linkedin.com/v2/me

X/Twitter

1. [Twitter Developer Portal](#)
2. App → “Elevated”
3. **OAuth 2.0 Client ID + Secret**
4. n8n → credential OAuth2 API





1. +

```
{
  "task": "生成一张关于 人工智能 的图片",
  "platform": "instagram",
  "contentType": "image",
  "tone": "friendly",
  "publish": true
}
```

2.

```
{
  "task": "生成一张关于 人工智能 的图片",
  "platform": "linkedin",
  "contentType": "image",
  "tone": "professional",
  "publish": true
}
```

3.) (

```
{
  "task": "生成一张关于 猫咪 的图片",
  "platform": "instagram",
  "contentType": "image",
  "publish": false
}
```

4.) prompt (

```
{
  "task": "生成一段关于 猫咪 的视频",
  "platform": "tiktok",
  "contentType": "video",
  "publish": false
}
```



node “7a. Generate Image”:

- dall-e-3 () —
- dall-e-2 —

1. node HTTP Request
2. 8. Route Publisher
- 3.

node DALL-E:

- 1024x1024 ()
- 1792x1024 ()
- 1024x1792 ()



(Composite Score)

Relevance	30%
Authority	25%
Freshness	15%

Table 5 – continued

Business Fit	15%
Performance Relevance	15%

Built with 🧠 AGENTIC-RAG-Ω + 🌈 DALL-E + 📱 Social APIs + ⚡ n8n